

GAUTAM AGARWAL

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EDUCATION

Columbia University | New York [Dec 2026]
Master of Science Computer Science - Machine Learning Track GPA: 4.0/4.0
Courses: NLP, High Performance ML, GenerativeAI, AgenticAI Systems, Big Data, StartUp Studio
Certificates: ISSO Engage, ISSO Global Leadership, Columbia StartMeUp Bootcamp, PDL Fellowship, PDS Scholarship

University of Wisconsin-Madison | Madison, Wisconsin [Dec 2022]
Bachelor of Science - Computer Science & Data Science GPA: 3.96/4.0
Awards: L&S Scholarship Fall 2021, Summer Study Scholarship 2020, Summer Internship Scholarship 2020, Dean's List

SKILLS & LANGUAGES

- **Languages:** Python, Java, R, SQL, C, C++, C#, M, HTML/CSS, JavaScript, PHP, React, Kotlin, Dart, MATLAB, Octave
- **Cloud Platforms:** AWS (Lambda, EC2, Sagemaker, Cloudwatch, API Gateway, MQ), GCloud (Compute Engine, AI, Apigee), Azure
- **Tools & Tech:** Terraform, Flutter, Unity, GitLab, Arduino, SonarQube, SDVI, Flask, sci-kit learn, MastML, Google Internal Tools

PROFESSIONAL EXPERIENCES

Software Engineering Intern, Google Photos - Modelling [May 2026 - Sep 2026]

- Engineered 4 mathematical post-processing methodologies to mitigate over-segmentation in the [Wardrobe](#) pipeline, eliminating user visible duplicate items and reducing LLM API calls by 44% in benchmark testing
- Conducted 8 prompt tuning experiments to engineer alternative methods for improving precision on hard-to-cluster apparel.
- Developed internal web-based prototyping environment to evaluate these LLM-based strategies on authentic data.
- Migrated model evaluation framework from research notebooks to production-grade distributed system for automated testing.

Lead Graduate Teaching Assistant, Columbia Engineering [Jan 2026 - Present]

- Directed teaching team for High Performance Machine Learning course with 120+ students, coordinating staff operations and curriculum planning
- Designed and graded technical assignments covering distributed training, model optimization, and GPU acceleration techniques
- Mentored students through office hours and individual consultations, translating complex ML systems concepts into practical implementations

Machine Learning Engineering Intern, ArklexAI [Aug 2025 - Dec 2025]

- Architected multi-turn conversational AI framework to transition from isolated prompt requests to system-user dialogue patterns, enabling contextually-aware GenAI interactions at scale.
- Extended global language support across core platform, improving accessibility and enabling deployment across diverse international customer bases.

Software Engineer, Epic Systems, Epic Assistant - Natural Language Platform [Mar 2023 - Aug 2025]

- Created general-purpose NLP library to support user interactions in 6 Epic applications; reduced LLM dependency by 80%
- Set up LLM integration for conversational AI chatbot platform to automate entity extraction
- Designed and constructed dynamic backend infrastructure for chatbot to extend functionality for 3 Epic applications
- Led research project to enhance LLM output for multiple chatbot functionalities, improving automation by 76%
- Introduced native NLP in search snippets, improving user response by 62%
- Developed pre-charting clinical summarization tool cutting down provider prep time by 25 minutes per patient
- Directed investigation into building native Speech-To-Text service (STT) to deploy across organization
- Designed Chatbot - agentic framework integration, unlocking orchestration features and extending AI adoption across 6 products
- Stretched organization roadmap by pitching ideas to enhance end-user experience across 5 AI-based features
- Mentored 4 interns; provided expertise in STT & LLM-focused engineering for TranslationalAI, ConversationalAI & Summarization

- Engaged with customers as product expert during multiple customer conferences and group calls with 750 customers
 - Judged 3 organization-sponsored hackathons and contributed to 4 university recruiting events, engaging with 400+ students
- Software Engineering Intern, Google - Display & Video 360** [May 2022 - Sep 2022]
- Developed workflow using Python and Google tools to automate cost model generation and weekly update
 - Engineered fast, lightweight validation method to replace traditional iterative A/B validation method
 - Decreased 7 manual steps to a single confirmation step; reduced runtime from 16 - 28 hours to 8 hours; decreased server load
- Media Workflow Engineering Intern, WarnerMedia** [Sep 2021 - Apr 2022]
- Developed media workflow software using SDVI and AWS platforms
 - Participated in Scrum methodologies and teamwork strategies like pair programming & code reviews
- Enterprise Integration Intern, UW-Madison Division of Information Technology** [Feb 2021 - Sep 2021]
- Helped integrate IT systems and data across the UW, worked with modern DevOps technologies and cloud platforms
 - Migrated systems integration to new Platforms: Informatica, AWS, & Apigee
- Undergraduate Teaching Assistant, UW-Madison Dept. of Computer Science** [Jan 2021 - May 2021]
- Designed debugging exercises, projects, and discussion content for Python course exploring web scraping and machine learning
 - Guided up to 375 students throughout the course during office hours and weekly group mentoring sessions
- AI & Computer Vision Intern, CamfyVision Innovations** [May 2020 - Sep 2020]
- Developed products through Data Collection, Inference Engine Coding, Model Training & Testing to cater to client needs
 - Products: Face Mask Detection, Social Distance Detection, Employee Detection, Solar Cell Detection, Helmet Detection
- Project Intern, The Tata Power Company Ltd. | Mumbai, India** [Jul 2018 - Oct 2018]
- Designed a PHP-based web application to monitor hydroelectric dam health remotely and store analyzable data
 - Displayed various parameters including flow rate, reservoir volume, and gate integrity on a dashboard

RESEARCH PROJECTS

- Self-Adaptive self-learning GenerativeAI Systems** [Sep 2025 - Present]
- Architected closed-loop self-improvement framework integrating LLM-based error detection, automated fine-tuning orchestration, and adaptive scheduling, enabling generative models to autonomously improve post-deployment without manual retraining.
 - Implemented adaptive hyperparameter optimization pipeline with performance-driven scheduling algorithm that dynamically adjusts fine-tuning frequency, mitigating catastrophic forgetting while maximizing convergence efficiency and cost reduction.
 - Demonstrated generalization across domains using Speech-to-Text as primary use case, achieving 10-20% relative WER improvements while maintaining platform abstraction for extensibility to text generation, translation, and other generative tasks.
- Agentic Migration of Clinical Chatbots** [Jan 2026 - Present]
- Designed and led migration of clinical chatbot towards an agentic infrastructure
 - Led research study comparing effects of chatbot performance between agentic infrastructure and monolithic prompt structure
- NFR Benchmarking Framework for IT Automation Agents** [Sep 2025 - Feb 2026]
- Conducted comprehensive literature review synthesizing 182+ sources to establish unified NFR taxonomy bridging software quality standards with agent-specific observability requirements, identifying critical gaps in production-readiness evaluation
 - Implemented streaming API support for ITBench CISO agents using GPT-4o-mini, instrumenting Langfuse-based telemetry to capture Time-to-First-Token (0.79-1.69s) and token generation speed (59-72 tokens/s), revealing prefill latency as the dominant bottleneck for interactive security workflows
 - Extended ITBench evaluation infrastructure with performance metrics enabling quantitative comparison of ReAct vs. PlanExecute architectures across latency, cost efficiency, and reliability dimensions
- On-Policy Knowledge Distillation for Speculative Decoding in LLMs** [Sep 2025 - Dec 2025]
- Investigated divergence measures (Forward KL, Reverse KL, JSD) for on-policy distillation to optimize draft-target alignment in speculative decoding, demonstrating 4.4% improvement in token acceptance rate for mathematical reasoning tasks
 - Led experimentation across Llama and SmoLLM model families, implementing white-box token-level distillation on GSM8K and CNN-DailyMail datasets to validate task-dependent divergence superiority

- Engineered dynamic batched speculative decoding evaluation framework with jagged sequence handling, enabling rigorous entropy-aware analysis revealing mode-seeking advantages in low-entropy tasks and robustness of JSD in high-entropy regimes

Machine Learning To Identify Diabetic Pathways in Mice and Humans [\[Link\]](#) [May 2021 - Dec 2022]

- Led team representing university in international ML **research** competition, competing against 21 universities worldwide
- Applied 16 ML and Data Science Methods to correlate insulin & glucose levels in mice with gene expression data
- Extracted expressive genomes from models using ExplainableAI solutions and extrapolated results with human genomes

Machine Learning for Air Quality Prediction from Wildfires [\[Link\]](#) [Aug 2020 - Apr 2021]

- Led the team representing the university in an international machine learning research competition.
- Compared deep learning methods to predict the air quality impact of wildfires in Northern California.
- Created data processing pipelines to fetch, clean, complete & combine data from NASA satellites and NOAA ground stations.
- Analyzed data to establish patterns between vegetation, humidity, air quality, and other factors around wildfires.

Water Infrastructure [\[Link\]](#) [Sep 2020 - May 2021]

- Used data analysis and machine learning to optimize the maintenance of roads and water mains in Madison
- Analyzed breakage factors and predicted future breakages in water pipes
- Factored road quality to suggest a combined repair plan

Uncertainty Estimation [\[Link\]](#) [Jan 2021 - Aug 2021]

- Used error analysis to estimate the domain of standard machine learning models
- Explored the relationship between different distance metrics and the error in different error metrics to establish blind spots.

SOFTWARE PROJECTS

MisfitsAI [Jan 2026 - May 2026]

- Led product discovery and roadmap planning for MisfitAI Wardrobe Planner, mobile app that turns users' closet and calendar into daily outfit recommendations for busy students and young professionals.
- Conducted structured customer discovery interviews and translated insights into onboarding flows, wardrobe capture requirements, and objection handling that informed landing page copy and in product messaging.
- Partnered with engineers to scope and ship the first recommendation system MVP, defining API contracts, edge cases, and evaluation criteria for calendar and wardrobe based outfit suggestions

Rx.AI [Apr 2026 - May 2026]

- Built dynamic AI-powered patient check-in system using that generates personalized clinical questionnaires
- Integrated voice and camera multimodal features including TTS question playback, real-time STT answer transcription), and inline image capture for wound photos and medication labels
- Designed end-to-end evaluation pipeline using DeepEval metrics with a BigQuery data sink to assess STT, TTS, image analysis, and question generation quality across 5 workflow combinations

Nutrigraph [Jan 2026 - Mar 2026]

- Built agentic multimodal RAG system that estimates dish macros from images and text using USDA and OpenFoodFacts.
- Designed diner and restaurant workflows, including restaurant search, dish logging, nutrition dashboards, and catalog creation.
- Added evaluation and feedback loops for macro accuracy, confidence based clarification, and repeated request handling.

TaskflowAI [Sep 2025 - Jan 2026]

- Engineered end-to-end agentic workflow system to parse email, intelligently extract tasks and calendar events, categorize by user preferences, and auto-populate productivity tools, reducing manual scheduling overhead.
- Designed orchestration layer for multi-agent coordination, enabling seamless integration between task extraction, categorization, and calendar management components.

MeetFree [\[Link\]](#) [May 2024 - Jun 2024]

- Built web application to record and auto-summarize meetings, simplifying manual note-taking for team discussions
- Leveraged Google Cloud to establish infra and engineered transcription processing multi-speaker audio via Google Speech-To-Text
- **Engineered** prompt-based summarization service utilizing Gemini to generate concise, structured notes, cutting review time

Cybersecurity Jacket

[Sep 2022 - Dec 2022]

- End-to-end Software engineering solution to demonstrate cybersecurity awareness among university students.
- Jacket with rubber ducky that connects to a computer, analyzes security and displays results on the jacket through self-sewn lights
- Developed backend infrastructure for Arduino: VM communication, web integration, and client-to-jacket light data translation

Blind Man's Bluff [\[Link\]](#)

[HackMIT 2020]

- Built a mobile application in Flutter to help people who are blind navigate through their daily routines.
- Used state-of-the-art real-time object detection and distance estimation to alert the user with vibrations & audio feedback.

COVID Bot [\[Link\]](#)

[SOS Hackathon 2020]

- Built an automated mopping robot to sanitize surfaces in hospitals during the COVID-19 outbreak
- Used Arduino and object detection algorithms to maneuver the robot and Ultraviolet light to sanitize surfaces.

LEADERSHIP

President, The Data Science Club [\[Link\]](#)

[Sep 2020 - Dec 2022]

- Spearheaded 950-member organization to help students transform interests into career opportunities
- Organized networking events, research seminars, and career development & practical learning workshops
- Created collaboration opportunities through hackathons & group projects; secured industry sponsorships for student support

Co-Founder & Vice-President, Badger Cricket Club [\[Link\]](#)

[Nov 2019 - Dec 2022]

- Built a community of 350+ cricket enthusiasts through inclusive organizational efforts.
- Advocated and secured athletic department recognition, funding, and institutional support for the club.
- Managed national organization affiliation, taught cricket, and organized matches, live streams, & networking events.

Vice-President, Undergraduate Computing Council

[Jan 2022 - Dec 2022]

- Established a council to bridge the gap between students and university professionals.
- Advocated for aligning classroom education with evolving industry requirements to enhance student readiness.
- Spearheaded diversity initiatives to empower underrepresented students and foster inclusivity.